

BOX : Homework 02

Question 1.

N/A

Question 2-4.

File	Number of sequences	Length (bases)	Species	Number of canonical 20-mers
file1.fa	1	2.22×10^6	Streptococcus Pneumoniae	2.15×10^6
file2.fa	1	2.99×10^4	SARS-CoV-2	2.99×10^4
file3.fa	1	9.18×10^3	HIV	9.08×10^3
file4.fa	7	2.55×10^6	Streptococcus	2.27×10^6
file5.fa	1	2.97×10^4	SARS-CoV-2	2.85×10^4
file6.fa	1	2.13×10^5	Streptococcus	1.96×10^5

Question 3.

While the fourth and sixth file both contain sequences from the Streptococcus genus, I could not identify exactly which species they were from.

Question 4.

I made the decision to replace all ambiguities, assigning a fixed base to each ambiguity (eg. “Y” is always replaced by “C”).

Calculating the Jaccard indexes for each pair of files yields the following table :

	file1.fa	file2.fa	file3.fa	file4.fa	file5.fa	file6.fa
file1.fa	1	1.85×10^{-6}	0	4.28×10^{-3}	9.27×10^{-7}	3.57×10^{-5}
file2.fa	1.85×10^{-6}	1	0	0	8.18×10^{-1}	0
file3.fa	0	0	1	0	0	0
file4.fa	4.28×10^{-3}	0	0	1	0	8.92×10^{-2}
file5.fa	9.27×10^{-7}	8.18×10^{-1}	0	0	1	0
file6.fa	3.57×10^{-5}	0	0	8.92×10^{-2}	0	1

We notice that the table is symmetrical ($J(A, B) = J(B, A)$, so no surprises there) and that the diagonal is made of ones ($J(A, A) = 1$, again, no surprises).

What is more interesting to note is that file 3 has no similarity to any other file (being the only instance of HIV genome) and that files 2 and 5 have a lot of similarities to each other (as they are both genomes of SARS-CoV-2) but are very different from all other files.

Finally, we can see that files 1, 4 and 6 all have some amount of similarities to one another, but not nearly as much as between files 2 and 5. This is likely because, while they all come from members of the Streptococcus genus, they do not come from the same species within that genus.

Question 5.

Now, A is represented by 00, C by 01, G by 10, and T by 11.